

ATUL KUMAR

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Profile

ECE student with strong foundation in C++, DSA, and IoT development. Experienced in ESP32-based embedded systems and smart automation projects. Interested in software development and embedded systems. Quick learner with good communication and teamwork abilities.

Education

Godavari Institute of Engineering and Technology 2022 – 2026

B.Tech in Electronics and Communication Engineering (CGPA: 7.5)

Govt Higher Secondary School, Garh Dhamma 2022

12th (80%)

Experience

Intern – Embedded Systems / IoT Feb 2026 to May 2026

Genz Educate Wing

- Developed the Adaptive Smart Traffic Signal Controller project as part of the internship, implementing FSM-based signal control logic using ESP32.
- Worked on real-time embedded control, sensor integration, and simulation using Wokwi Simulator under mentorship.

Projects

Adaptive Smart Traffic Signal Controller

ESP32, Arduino IDE, Wokwi Simulator, OLED Display, LEDs, Push Buttons

- Designed FSM-based intelligent traffic control with adaptive signal timing based on real-time traffic density.
- Implemented emergency vehicle priority handling and pedestrian crossing management with countdown display.
- Built event logging and OLED-based traffic status monitoring for real-time decision making.

Smart Home Automation System

ESP32, Arduino IDE, PIR Sensor, LDR, DHT22, Relay Module, WiFi

- Built IoT-based system to automatically control home appliances using PIR, LDR, and DHT22 sensors.
- Implemented automatic lighting, motion-based security, temperature monitoring, and energy-efficient automation logic.

Green Energy Based Portable Charging Hub

Solar Energy, Power Electronics

- Designed a solar-powered portable charging system focused on sustainability and energy efficiency.

Skills

- **Programming:** C++, Python (Basics)
- **Embedded/IoT:** ESP32, Arduino IDE, Wokwi Simulator, Relay Modules, Sensors (PIR, LDR, DHT22)
- **Core CS:** Data Structures & Algorithms (DSA)
- **Tools:** Git (Basics), Arduino IDE
- **Soft Skills:** Problem Solving, Communication, Teamwork, Adaptability
- **Languages:** English, Hindi, Telugu

Certifications & Achievements

- Blockchain and Applications – Great Learning
- Prompt Engineering – Great Learning
- Python Programming – Great Learning
- Cloud Fundamentals – Skill Camp
- Head Boy, GHSS Garh Dhamma (2020–2021)
- PMSSS Scholarship, Academic Excellence (2021)
- Completed 12th with 80% from Govt Higher Secondary School, Garh Dhamma